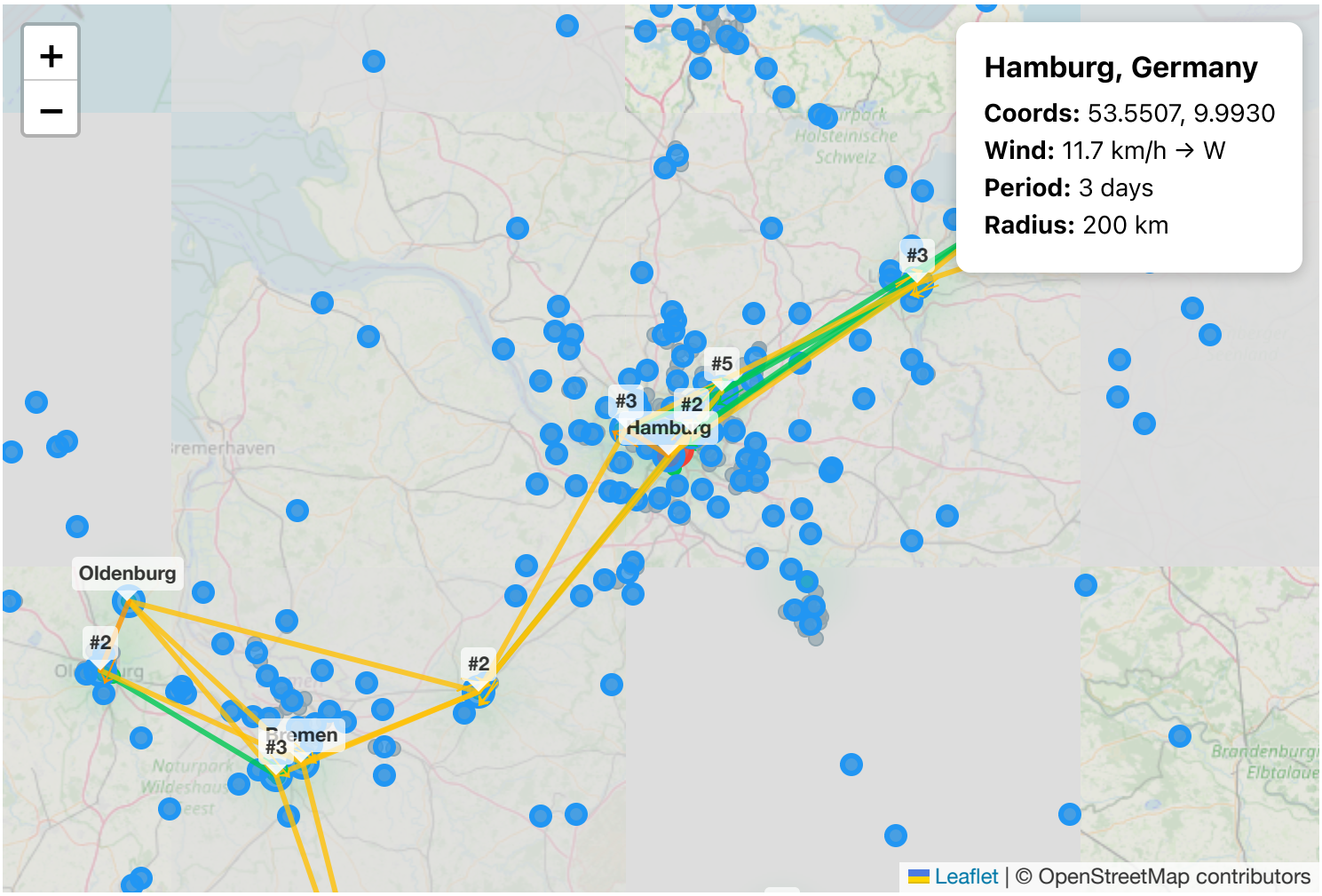

Air Quality Report — Hamburg, Germany

Key Insights

- **Highest PM2.5:** 35.4 µg/m³ (AQI 100, Moderate) at Hamburg (6 sensors) — 2026-03-14 19:00
- **Strongest front:** Lubeck-2 → Hamburg-2 (11 sensors) at 7 km/h SW (correlation 100%, lag 11h)
- **Wind:** 11.7 km/h W — moderate — pollution disperses gradually
- **Assessment:** Moderate air quality. Generally safe, sensitive individuals may want to limit prolonged outdoor exertion.

Spikes

City	Time	PM2.5	Delta	Z-score
Minden (2 sensors)	2026-03-15 04:00	8.4	+2.8	3.1
Minden (2 sensors)	2026-03-15 05:00	10.2	+1.8	2.0

City	Time	PM2.5	Delta	Z-score
Hamburg (6 sensors)	2026-03-14 17:00	34.5	+26.6	3.2
Hamburg (6 sensors)	2026-03-14 19:00	35.4	+23.9	2.9
Hannover	2026-03-15 06:00	7.6	+2.2	2.5
Hannover	2026-03-15 07:00	9.6	+2.0	2.2
Hamburg-2 (11 sensors)	2026-03-14 22:00	8.6	+4.3	3.8
Oldenburg	2026-03-14 10:00	6.7	+5.9	3.1
Oldenburg	2026-03-14 11:00	15.7	+9.1	4.7
Oldenburg	2026-03-15 10:00	5.7	+3.9	2.0
Hamburg-3 (2 sensors)	2026-03-14 07:00	5.9	+2.5	2.1
Hamburg-3 (2 sensors)	2026-03-15 13:00	8.1	+3.0	2.5
Wolfsburg	2026-03-15 03:00	6.9	+3.6	2.1
Wolfsburg	2026-03-15 04:00	11.8	+4.9	3.0
Wolfsburg	2026-03-15 19:00	10.6	+4.6	2.8
Osnabruck	2026-03-14 22:00	7.3	+2.6	3.5
Bremen	2026-03-15 13:00	8.7	+5.9	2.0
Lubeck-2	2026-03-15 07:00	11.8	+7.8	3.6
Lubeck-2	2026-03-15 18:00	7.6	+5.0	2.3
Osnabruck-2	2026-03-14 03:00	8.1	+6.5	2.1
Osnabruck-2	2026-03-14 17:00	22.2	+11.4	3.7
Wolfsburg-2	2026-03-15 01:00	10.1	+3.7	2.1
Wolfsburg-2	2026-03-15 08:00	19.3	+4.6	2.6
Lubeck-3 (2 sensors)	2026-03-15 17:00	5.7	+3.0	2.8
Bremen-2	2026-03-15 04:00	19.0	+5.8	2.1
Bremen-2	2026-03-15 10:00	19.9	+8.0	3.0
Hamburg-5 (2 sensors)	2026-03-14 20:00	4.6	+1.6	2.6
Hamburg-5 (2 sensors)	2026-03-15 14:00	4.5	+1.4	2.4
Oldenburg-2 (6 sensors)	2026-03-14 09:00	12.7	+4.0	2.3

City	Time	PM2.5	Delta	Z-score
Rostock	2026-03-15 05:00	12.7	+3.8	2.6
Rostock	2026-03-15 11:00	5.7	+3.5	2.4
Bremen-3 (2 sensors)	2026-03-14 09:00	15.4	+5.6	2.3
Bremen-3 (2 sensors)	2026-03-15 00:00	16.9	+6.4	2.6

Pollution Fronts

From → To	Speed (km/h)	Direction	Lag (h)	Correlation
Lubeck-2 → Hamburg-2 (11 sensors)	7	SW	11	100%
Lubeck-2 → Hamburg-5 (2 sensors)	5	SW	12	100%
Lubeck-2 → Rostock	3	E	14	96%
Hamburg-2 (11 sensors) → Hamburg-5 (2 sensors)	10	NE	1	95%
Hamburg-5 (2 sensors) → Lubeck-3 (2 sensors)	15	NE	3	93%
Hamburg-2 (11 sensors) → Lubeck-3 (2 sensors)	14	NE	4	93%
Hamburg-3 (2 sensors) → Hamburg-2 (11 sensors)	13	E	1	91%
Hamburg-3 (2 sensors) → Hamburg-5 (2 sensors)	21	E	1	90%
Wolfsburg → Hannover	21	W	3	88%
Hamburg-5 (2 sensors) → Bremen-2	40	SW	2	88%
Minden (2 sensors) → Hannover	37	E	1	87%
Oldenburg-2 (6 sensors) → Bremen-3 (2 sensors)	42	SE	1	86%
Bremen → Minden (2 sensors)	18	S	4	85%
Hamburg-2 (11 sensors) → Bremen-2	35	SW	2	84%
Lubeck-2 → Lubeck-3 (2 sensors)	2	SW	12	84%
Oldenburg → Bremen-3 (2 sensors)	3	SE	14	83%

From → To	Speed (km/h)	Direction	Lag (h)	Correlation
Rostock → Lubeck-3 (2 sensors)	65	W	1	81%
Bremen → Bremen-2	20	E	2	81%
Hamburg-3 (2 sensors) → Lubeck-3 (2 sensors)	13	NE	5	80%
Bremen-3 (2 sensors) → Minden (2 sensors)	24	S	3	79%
Hamburg (6 sensors) → Hamburg-2 (11 sensors)	1	NE	7	78%
Wolfsburg → Wolfsburg-2	10	NW	3	77%
Hamburg-3 (2 sensors) → Bremen-2	21	SW	3	77%
Oldenburg-2 (6 sensors) → Bremen	46	SE	1	76%
Hamburg (6 sensors) → Lubeck-3 (2 sensors)	5	NE	12	76%
Oldenburg → Bremen	3	SE	14	75%
Osnabruck → Minden (2 sensors)	12	E	6	75%
Oldenburg → Bremen-2	4	E	17	75%
Hamburg (6 sensors) → Hamburg-5 (2 sensors)	2	NE	9	74%
Bremen-3 (2 sensors) → Bremen-2	45	E	1	73%
Hamburg (6 sensors) → Bremen-2	6	SW	10	73%

Methodology

Data sources: Open-Meteo atmospheric model (~11km grid) + Sensor.Community citizen science network (516 sensors, 301 analysis nodes).

Spike detection: Z-score on hourly PM2.5 differences. A spike is flagged when the rate of change exceeds 2 standard deviations above the rolling mean — indicating a sudden pollution event rather than normal variation.

Front tracking: Cross-correlation analysis between city pairs with time lags from -24h to +24h. The lag at peak correlation reveals how many hours pollution takes to travel between two points. Combined with haversine distance, this gives front speed and direction.

Confidence: When both Open-Meteo (model) and Sensor.Community (ground sensors) agree on lag direction, confidence is boosted. Disagreement reduces confidence, highlighting uncertain fronts.

AQI Reference

AQI	Category	Health Guidance
0–50	Good	No restrictions
51–100	Moderate	Sensitive individuals: limit prolonged outdoor exertion
101–150	Unhealthy for Sensitive	Active children and adults: reduce prolonged outdoor exertion
151–200	Unhealthy	Everyone: reduce outdoor exertion
201–300	Very Unhealthy	Everyone: avoid outdoor activity
301–500	Hazardous	Stay indoors. Close windows.

Generated by airq — open source CLI air quality analyzer